# Positive $C_p$-homeomorphisms and positive Banach isomorphisms coincide for countable compacta

Candidate full solution packet for arXiv:2601.11463, Question 4.3

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**Status.** `candidate full, likely valid; requires human review.`

## 1 Source question

Cúth, Havelka, Rondoš, and Sarı ask the following in Question 4.3 on page 27 of [1].

$$Sh(1) = h(\omega), \quad Sh(n+1) = h(n) - h(\omega), \quad n \in \mathbb{N}.$$

Another question related to our research concerns spaces of continuous functions with the pointwise topology. We recall that the classification of spaces of the form $\mathcal{C}_p(K)$, where $K$ is a countable compact space, by linear homeomorphism, coincides with the classification of the Banach spaces $\mathcal{C}(K)$ by linear isomorphisms, see [3]. Thus, the following problem seems natural.

**Question 4.3.** *Assume that $K, L$ are countable compact spaces. Is it true that there exists a positive linear homeomorphism of $\mathcal{C}_p(K)$ onto $\mathcal{C}_p(L)$ iff there exists a positive isomorphism of $\mathcal{C}(K)$ onto $\mathcal{C}(L)$?*

In transcription:

> Assume that $K, L$ are countable compact spaces. Is it true that there exists a positive linear homeomorphism of $C_p(K)$ onto $C_p(L)$ if and only if there exists a positive isomorphism of $C(K)$ onto $C(L)$?

As in [1], "positive" is one-sided: the displayed map is positive, but its inverse is not required to be positive.

## 2 Definitions

For compact $K$, the vector space $C(K)$ consists of the continuous real-valued functions on $K$. We write $C_p(K)$ for this vector space with the topology of pointwise convergence, and $C(K)$ for the Banach space with the supremum norm. (The linear-topological argument is unchanged over the complex scalars.)

For $x \in K$, let $\mathrm{ev}_x(f) = f(x)$. We call a linear map $T : C(K) \to C(L)$ *coordinate-finite* if, for every $y \in L$, there are finitely many $x_1, \ldots, x_r \in K$ and scalars $a_1, \ldots, a_r$ such that

$$(Tf)(y) = \sum_{j=1}^{r} a_j f(x_j) \qquad (f \in C(K)). \tag{1}$$

For infinite countable compact $K$, $\mathrm{ht}(K)$ denotes its Cantor–Bendixson height. Following [1], $\Gamma(\alpha)$ is the least ordinal of the form $\omega^\xi$ which is at least $\alpha$.

## 3 Proof intuition

The two topologies have complementary automatic-continuity mechanisms. Uniform convergence is stronger than pointwise convergence. Therefore, if a linear $C_p$-homeomorphism and a candidate uniform limit are placed in a closed-graph test, pointwise continuity identifies the limit. The closed graph theorem then upgrades the map and its inverse to bounded Banach-space operators.

The reverse implication is not automatic for an arbitrary positive bounded operator: its value at a point of $L$ could be integration against a measure with infinite support, which is not a continuous functional for the pointwise topology. What resolves the problem is the particular constructive proof in the source paper. Its positive isomorphisms and their inverses are coordinate-finite. This is exactly the missing $C_p$ continuity, and the property survives all the finite compositions and homeomorphic conjugacies in the source classification proof.

## 4 Elementary topology lemmas

**Lemma 1.** *A linear map $T : C(K) \to C(L)$ is continuous from $C_p(K)$ to $C_p(L)$ if and only if it is coordinate-finite.*

*Proof.* If (1) holds for every $y$, then every coordinate $\mathrm{ev}_y \circ T$ is continuous for the pointwise topology, which is precisely the continuity of $T$ into the product topology on $C_p(L)$.

Conversely, fix $y \in L$. The linear functional $\mathrm{ev}_y \circ T$ is continuous on $C_p(K)$. Hence there are $x_1, \ldots, x_r \in K$ and $\varepsilon > 0$ such that

$$|f(x_j)| < \varepsilon \ (1 \le j \le r) \quad \Longrightarrow \quad |(Tf)(y)| < 1.$$

If $f(x_j) = 0$ for all $j$, the same implication applies to every scalar multiple of $f$, so $(Tf)(y) = 0$. Thus $\mathrm{ev}_y \circ T$ factors through the finite-dimensional coordinate map $f \mapsto (f(x_1), \ldots, f(x_r))$. A linear functional on its range extends to $\mathbb{R}^r$, giving (1). □

**Lemma 2.** *Coordinate-finite linear maps are closed under finite composition. A homeomorphism $\phi : L \to K$ induces a coordinate-finite isometry $f \mapsto f \circ \phi$, whose inverse is also coordinate-finite.*

*Proof.* For composition, substitute each finite evaluation formula for the inner map into the finite formula for the outer map; only a finite union of finite supports occurs. A composition operator has exactly one input evaluation in each output coordinate, and the same is true for its inverse. □

**Lemma 3.** *If $T : C_p(K) \to C_p(L)$ is a linear homeomorphism, then the same algebraic map is a Banach-space isomorphism $T : C(K) \to C(L)$.*

*Proof.* Suppose $f_n \to f$ uniformly on $K$ and $Tf_n \to g$ uniformly on $L$. Uniform convergence implies pointwise convergence. Continuity of $T$ for the pointwise topologies gives $Tf_n \to Tf$ pointwise, while the assumed uniform convergence gives $Tf_n \to g$ pointwise. Hence $g = Tf$. The norm graph of $T$ is closed, so the closed graph theorem makes $T$ norm-bounded. Applying the same argument to the $C_p$-continuous inverse makes $T^{-1}$ norm-bounded. □

## 5 The coordinate-finite content of the source construction

We isolate the observation not stated in [1].

**Proposition 4.** *Let $K, L$ be infinite countable compact spaces satisfying*

$$\mathrm{ht}(K) \leq \mathrm{ht}(L) < \Gamma(\mathrm{ht}(K)). \tag{2}$$

*Then the constructive proof of Theorem 2.4 in [1] yields a positive isomorphism $U : C(K) \to C(L)$ for which both $U$ and $U^{-1}$ are coordinate-finite.*

*Proof.* We inspect all types of maps used in that proof. By the Mazurkiewicz–Sierpiński representation employed there, the spaces are first conjugated by homeomorphisms to ordinal intervals. Lemma 2 handles these conjugacies.

The first nontrivial map is the operator in Proposition 2.12 of [1]. For a fixed finite $k$, its formula for $(Tf)(y)$ is either one evaluation, a weighted sum of $k$ endpoint evaluations, or

$$(Tf)(\gamma) = \sum_{j=i}^{k} \lambda_j f(p_{i,j}(\gamma)) + \sum_{j=1}^{i-1} \lambda_j f(\omega^\alpha j) \qquad (\gamma \in I_i).$$

Thus $T$ is coordinate-finite. The displayed inverse in that proof is also coordinate-finite. At the finitely many endpoints this is immediate; on the last block it is a finite sum. On a block $J_l$ with $l < k$, the inductive formula is simplified in the source proof to

$$\begin{aligned} \lambda_l (Sh)(\gamma) = h(p_{l,l}^{-1}(\gamma)) + \lambda_l h(\omega^\alpha + l) \\ - h\Big(p_{l+1,l+1}^{-1}\big(p_{l,l+1}(p_{l,l}^{-1}(\gamma))\big)\Big). \end{aligned} \tag{3}$$

So each inverse output coordinate uses only three input evaluations.

The second nontrivial map is Proposition 2.15 of [1]. Its tuple length $\ell(\gamma)$ is bounded by a fixed finite integer. The operator is given by

$$\begin{aligned} (Tf)(*) &= f(\omega^\alpha), \\ (Tf)(\gamma) &= \left(1 - \sum_{i=1}^{\ell(\gamma)} \lambda_i\right) f(\omega^\alpha) + \sum_{i=1}^{\ell(\gamma)-1} \lambda_i f(\rho(\gamma|_i + 1)) + \lambda_{\ell(\gamma)} f(\rho(\gamma)). \end{aligned} \tag{4}$$

Every coordinate is a finite sum. More decisively, the inverse is explicitly

$$\begin{aligned} (Sh)(\omega^\alpha) &= h(*), \\ (Sh)(\rho(\gamma)) &= h(*) + \lambda_1^{-1}(h(\gamma) - h(*)) && (\ell(\gamma) = 1), \\ (Sh)(\rho(\gamma)) &= h(*) + \lambda_{\ell(\gamma)}^{-1}\big(h(\gamma) - h(\gamma|_{\ell(\gamma)-1} + 1)\big) && (\ell(\gamma) > 1). \end{aligned} \tag{5}$$

Hence both directions of this map are coordinate-finite.

Finally, the proof of Theorem 2.4 in [1] obtains the desired positive isomorphism by a finite chain consisting of the maps in Propositions 2.12 and 2.15 and homeomorphism-induced isometries. Lemma 2 therefore gives the asserted coordinate-finiteness of $U$ and $U^{-1}$. Positivity of $U$ is exactly the positivity verified in the source construction. □

# 6 Full answer

**Theorem 5.** *Let $K, L$ be countable compact spaces. The following are equivalent:*

(i) *there is a positive linear homeomorphism $T : C_p(K) \to C_p(L)$;*

(ii) *there is a positive Banach-space isomorphism* $S : C(K) \to C(L)$.

*If* $K, L$ *are infinite, these are further equivalent to*

$$\mathrm{ht}(K) \leq \mathrm{ht}(L) < \Gamma(\mathrm{ht}(K)). \tag{6}$$

*Thus the answer to Question 4.3 of [1] is affirmative.*

*Proof.* Assume (i). Lemma 3 makes $T$ and $T^{-1}$ bounded in the supremum norms. The algebraic map is already bijective and positive, so it is the positive Banach isomorphism required in (ii).

Conversely, first suppose $K, L$ are infinite and (ii) holds. Theorem 2.4 of [1] says that (ii) is equivalent to (6). Proposition 4 then produces a positive Banach isomorphism $U$ for which both $U$ and $U^{-1}$ are coordinate-finite. Lemma 1 says exactly that these two maps are continuous for the pointwise topologies. Hence $U : C_p(K) \to C_p(L)$ is the positive linear homeomorphism required in (i).

It remains only to dispose of finite cases. A finite compact Hausdorff space is discrete and $C(K)$ has dimension $|K|$. If one space is finite and the other infinite, neither kind of linear isomorphism can exist. If both are finite, a Banach-space isomorphism forces $|K| = |L|$; any bijection between them is a homeomorphism and its composition operator is a positive isometry and a $C_p$-homeomorphism. This proves the equivalence in all cases. □

# 7 Verification notes and limitations

- Lemma 3 does not use countability. Countability is used only to invoke the source classification and its ordinal construction in the reverse implication.
- We do not claim that every positive Banach isomorphism is $C_p$-continuous. We prove the existential statement by selecting the coordinate-finite isomorphism already constructed in the proof of [1, Theorem 2.4].
- The inverse need not be positive. Requiring positivity of both a map and its inverse would be a different order-isomorphism problem.
- No computational or unproved conditional dependency enters the proof. The only source-dependent audit is the finite-evaluation form of the explicit operators and inverses in Propositions 2.12 and 2.15 of [1].

# 8 Bounded novelty search

On 21 July 2026 we searched the run's lightweight indexes, the local 2000–2026 arXiv source corpus, the current arXiv record for [1], and arXiv web results. Queries included the exact phrase `positive linear homeomorphism C_p(K)`, the title and arXiv id `2601.11463`, all four authors' names, and close phrases involving positive $C_p$ classification of countable compacta. The current v1, dated 16 January 2026, still states Question 4.3. No later explicit answer was found. The returned July 2026 work on $C_p$ spaces over separable compact lines concerns a different classification, and a May 2026 source merely cites [1] in an operator-space context.

Novelty confidence is moderate rather than definitive. The implication is newly identified here, but its short proof relies on observing a property of the explicit formulas already printed earlier in the same source paper.

## 9 Human-review recommendation

Check the quantifier and orientation in the source proof of Theorem 2.4, then verify (3) and (5) directly against Propositions 2.12 and 2.15. Those checks establish that every elementary map and inverse is coordinate-finite. The remaining arguments are the elementary $C_p$ dual description, closure under composition, and the closed graph theorem.

## References

[1] M. Cúth, J. Havelka, J. Rondoš, and B. Sarı, *The classification of $C(K)$ spaces for countable compacta by positive isomorphisms*, arXiv:2601.11463v1 (2026).